MIL-G-38332A(USAF)
AMENDMENT 3
8 MAY 1987
SUPERSEDING
AMENDMENT 2
15 October 1982

MILITARY SPECIFICATION

GYROSCOPE, RATE SWITCHING CN-1050I/ASN-55

This amendment forms a part of MIL-G-38332A (USAF), dated 6 November 1969, and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1 Delete "first sentence" and add: "Government documents."

* Add: "2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

Under "Specifications, Federal", delete "QQ-P-416" and add the following:

"PPP-B-601 Boxes, Wood, Cleated Plywood"

Under "Specification, Military", add the following:

"MIL-C-83488 Coating, Aluminum, Ion Vapor Deposited"

Under "Standards, Military" change "MIL-STD-100" to read "DOD-STD-100".

Add the following boxed paragraph to bottom center of page:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* Under STANDARDS, Military, delete "MIL-STD-831".

* Add: "2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue

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of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race Street, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)"

* Add: "2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained."

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3.5.3.2 Delete entirely and add:

"3.5.3.2 Leak rate. The gyroscope shall be sealed so that the maximum leak rate shall not exceed 0.1 micron cubic foot per hour (0.787 micron cubic centimeter per second) at a pressure differential equivalent to 1 atmosphere."

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3.7 Line 3 delete "MIL-STD-100" and add: "DOD-STD-100".

PAGE 8

3.15.2 Delete entirely and add:

3.15.2 Steel parts. Steel parts shall be coated with vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the cognizant engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

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* Add: "3.18 Recycled and reclaimed materials. Recycled and reclaimed materials shall be used to the maximum extent possible without jeopardizing the end use of the item."

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* Add: "4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptnce comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

PAGE 10

* Delete "4.4.2" and add the following new paragraph:

"4.4.2 Test Records. Throughout all qualification testing called for in paragraph 4.4. the contractor shall maintain complete test records and keep them available for inspection upon request by the local government representative. As a minimum the records shall include the following IAW MIL-STD-781: Operating sheet, log of failures, failure reports, equipment repair sheet, failure tag and failed-part summary analysis."

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4.6.7: Delete entirely and add:

"4.6.7 Leak rate. The sealed case shall be tested for leak by means of a mass-spectrometer-type helium leak detector or other means approved by the contracting activity. The leak rate shall not exceed 0.1 micron cubic foot per hour (0.787 micron cubic centimeter per second) at a pressure differential of 1 atmosphere."

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4.6.16 Line 2 delete "method 2" and add: "Test Method 9".

* 4.7 Delete entirely and add: "4.7 Inspection of Packaging. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129"

* Delete Section 5 in its entirety and Add:

"5. PACKAGING

5.1 Preservation. Preservation shall be level A, C or Industrial, as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Cleaning. Gyroscope shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Gyroscope shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each gyroscope shall be packaged in quantity unit packs of one each in accordance with Method IIC of MIL-P-116. Each gyroscope shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the gyroscope when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each gyroscope shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of gyroscope shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Gyroscope packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistant with the protection required.

5.2.2 Level B. Gyroscope packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged gyroscope shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129."

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* Delete "6.4".

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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